

SARDAR PATEL UNIVERSITY, VALLABH VDIYANAGAR
BCA (4TH SEMESTER) (NC) EXTERNAL EXAMINATION

28TH MAY 2021, FRIDAY

Subject: US04FBCA01: Computer Based Numerical & Statistical Method

Time: 10:30a.m to 12: 30p.m

Total Marks:70

Note: Scientific Calculator is Allowed

Q.1. Multiple Choice Questions

[10]

1. Which method is not iterative method from following?
(a) False position (b) Bisection (c) ~~LaGrange's~~ (d) none of them
2. Which method is the best method to obtain root of equation $f(x)=0$. From the following?
(a) False position (b) Bisection (c) ~~Newton's Raphson~~ (d) none of them
3. The number 0.01850×10^3 has _____ significant digits
(a) 3 (b) 4 (c) 5 (d) ~~6~~
4. All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _____.
(a) ~~Polynomial~~ (b) Equation (c) Algorithm (d) None
5. _____ method is used if the estimated value lies towards the end of the difference table.
Divided difference (b) Forward difference (c) ~~Backward difference~~ (d) None
6. _____ method of interpolation is used for unequally spaced functions.
(a) Forward difference (b) backward difference (c) ~~Newton's divided difference~~ (d) None
7. The system of linear equation $AX = B$ is said to be homogeneous if...
(a) $A \neq 0$ (b) ~~$B = 0$~~ (c) $A = 0$ (d) A is symmetric
8. The system of linear equation $AX = B$ is said to be non-homogeneous if...
(a) ~~$B \neq 0$~~ (b) $B = 0$ (c) $A \neq 0$ (d) A is symmetric
9. Forecasts _____
a. become more accurate with longer time horizons
b. ~~are rarely perfect~~
c. are more accurate for individual items than for groups of items
d. all of the above
10. One purpose of short-range forecasts is to determine _____.
a. production planning
b. inventory budgets
c. facility location
d. ~~job assignments~~

Q.2 True False and Fill in the Blanks

[08]

1. If for a real continuous function $f(x)$, $f(a)f(b) < 0$, then in the range of $[a,b]$ for $f(x)=0$, there is at least one root
2. The second order differences is denoted by Δ^2
3. If population census for the years 1931, 1941, 1951, 1961 and 1971 is given and if we want to estimate the population for the year 1935 then forward diff method is used.
4. Rate of change of distance with respect to time represents speed
5. The system of linear equation $AX = B$ can be solved by matrix inversion method only if $|A| \neq 0$. State True or False
6. The sales of departmental store on Dushera and Diwali are associated with the component of a time series is seasonal variation. State True or False
7. Linear trend of a time series indicates towards constant rate of change. State True or False
8. The moving averages in a time series are free from the influences of seasonal and irregular variations. State True or False

CCDA CCBA BD

Q.3 Short Questions (Attempt any Ten)

[20]

- 1 Define the terms (1) Relative error and (2) absolute error
- 2 To obtain approximate solution of the equation $X^3 - 9x + 1 = 0$. (False Position Method)
- 3 To obtain approximate solution of the equation $2^x - x - 3 = 0$. (Successive Approximation Method)
- 4 Draw forward difference table.
- 5 Draw backward difference table.
- 6 Define Interpolation and Extrapolation
- 7 List only various direct and iterative methods
- 8 If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$?
- 9 If x lies in the lower half of the table and if $x_{k-1} < x < x_k$, then what is $\frac{dy(x)}{dx}$?
- 10 What is Time Series?
- 11 What do you mean by Secular trend?
- 12 Differentiate between linear and nonlinear trend.

Q.4 Long Question Answers (Attempt any four)

[32]

- 1 Bisection method Find significant root up to three decimal points
 $X^3 - 4X - 9 = 0$ *Root lies 2 and 3 2.687*

- 2 False Position Method. Find significant root up to three decimal points
 $X^3 - X - 1 = 0$ *0 and 1 0.757*

- 3 Inverse Interpolation

The value of x and $y = f(x)$ are given below

x	5	6	9	11
$f(x)$	12	13	14	16

Find value of x when $f(x) = 15$.

- 4 Given the table of values as (use Lagrangian method)

X	0	1	2	3
$y(x)$	0	2	8	27

Find $y(2.35)$.

- 5 Given the following table

x	0	1	2	3	4	5	6
y	0	2	2.75	3.4	3.2	2.5	2.2

Use backward difference formula to compute $\frac{dy}{dx}$ at 6.

- 6 Given the following table

x	0	1	2	3	4	5	6
y	0	2	2.75	3.4	3.2	2.5	2.2

Use forward difference formula to compute $\frac{dy}{dx}$ at 2.25

- 7 Obtain seasonal indices using simple average method.

Year	summer	monsoon	Autumn	Winter
1	30	81	62	119
2	33	104	86	171
3	42	153	99	221
4	56	172	129	255
5	67	201	136	302

- 8 Calculate three yearly moving averages for the following data

YEAR	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960
Y	242	250	252	249	253	255	251	257	260	265	262